Breast Cancer Classification Using Neural Network Architectures: From Linear Models to Convolutional Networks

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***Abstract*—This paper presents a comparative study of five neural network architectures applied to binary classification of breast tumors as benign or malignant, using the breast ultrasound image dataset from Al-Dhabyani et al. The study follows a progression from simple linear models to deep convolutional networks: Least Mean Squares (LMS), Pocket Perceptron, Multilayer Perceptron (MLP), a Convolutional Neural Network (CNN) trained on binary tumor masks, and a transfer learning approach using ResNet18 on raw ultrasound images. Each model was implemented from scratch where required and evaluated on the same test set using F1-Score, Precision, Recall, and Accuracy. Results show that the Pocket Perceptron (F1=0.9206) and MLP (F1=0.9508) achieve strong performance on handcrafted morphological features, the Mask CNN reaches F1=0.9677 by leveraging clean lesion shape information, while the raw-image ResNet18 is limited to F1=0.7812 due to speckle noise and the absence of explicit tumor localization. These findings highlight the critical role of input representation quality in medical image classification.**

***Index Terms*—breast cancer classification, neural networks, backpropagation, convolutional neural networks, transfer learning, medical image analysis**

I. LMS: Single-Layer Linear Classifier

A. Methodology

The Least Mean Squares (LMS) algorithm, introduced by Widrow and Hoff [1], was implemented as a single neuron with linear activation and MSE loss. Seven morphological features extracted from tumor masks (*Texture, Perimeter, Cx, Cy, Concave, Concave_Points, Smoothness*) were used as input. The dataset consists of 454 benign and 211 malignant segments, reflecting a natural class imbalance of approximately 2.15:1. Data was split into 70% training, 15% validation, and 15% test sets. Weight initialization used small random values, and predictions were thresholded at 0.5 for binary classification.

B. Hyperparameter Search

A grid search over learning rates $\eta \in \{0.01, 0.05, 0.1\}$ and epochs $\in \{50, 100, 200\}$ was conducted. Table I summarizes the F1-Score on the validation set for each configuration.

The best configuration ($\eta = 0.1$, epochs=200) was selected for final evaluation on the test set.

TABLE I
LMS Hyperparameter Search — F1-Score on Validation Set

η	Epochs	F1-Val
0.01	50	89.66%
0.01	100	89.66%
0.01	200	89.66%
0.05	50	89.66%
0.05	100	89.66%
0.05	200	91.53%
0.10	50	91.53%
0.10	100	91.53%
0.10	200	**91.53%**

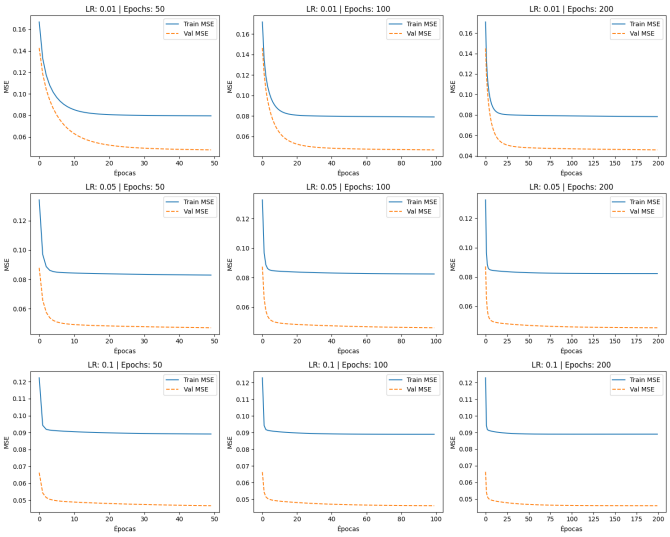

Fig. 1. Training and validation MSE for LMS across learning rates and epochs.

C. Results and Analysis

The dispersion matrix in Fig. 2 provides the foundation for understanding the model's behavior. Features such as *Texture*, *Perimeter*, and *Concave_Points* show partial class separation between benign (Class 0) and malignant (Class 1) tumors, suggesting that a linear boundary can capture a significant

portion of the class structure. However, features like *Cx*, *Cy*, and *Smoothness* exhibit substantial overlap, indicating that no single hyperplane can perfectly separate the full feature space.

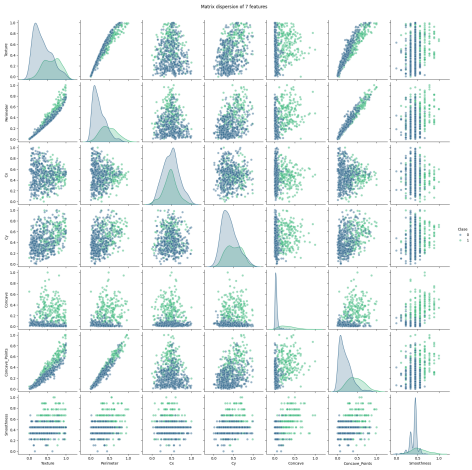

Fig. 2. Dispersion matrix of the 7 morphological features. Blue: benign (Class 0), green: malignant (Class 1). Diagonal shows per-class KDE distributions.

The error surface in Fig. 3 confirms a strictly convex MSE landscape over the weight space, with a single global minimum and no local minima. This convexity guarantees that gradient descent reliably converges to the optimal solution regardless of the chosen hyperparameters, which explains why the grid search over $\eta \in \{0.01, 0.05, 0.1\}$ and epochs $\in \{50, 100, 200\}$ produced negligible differences in MSE across all nine configurations (Table I).

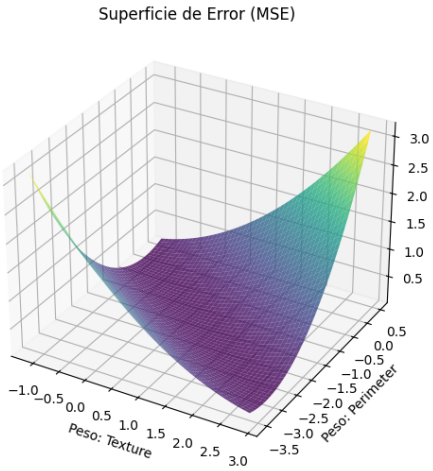

Fig. 3. MSE error surface as a function of two representative weights (*Texture* and *Perimeter*). The strictly convex shape guarantees convergence to the global minimum regardless of initialization or learning rate.

The final evaluation on the test set is summarized in Table II. The confusion matrix yielded TN=69, FP=0, FN=7, TP=25. Despite the simplicity of a linear boundary, the model achieved perfect Precision with zero false positives, confirming that the linearly separable portion of the feature space is reliably classified. However, the seven missed malignant cases (Recall=78.12%) reflect the fundamental limitation identified in the dispersion matrix: overlapping features prevent the linear boundary from capturing all malignant patterns. The minimal Train/Val MSE gap across all configurations further confirms the absence of overfitting, consistent with the low capacity of the model.

TABLE II
LMS FINAL TEST RESULTS

Metric	Value
Accuracy	93.07%
Precision	100.00%
Recall	78.12%
F1-Score	87.72%

II. PERCEPTRON: HARD-LIMIT DECISION BOUNDARY

A. Methodology

Two variants of the Perceptron [2] were implemented: the standard online update rule and the Pocket algorithm, which retains the best weight vector found during training. Both use a step activation function and the Delta rule for weight updates. Random weight initialization was applied to break symmetry.

B. Hyperparameter Search

Both variants were evaluated across $\eta \in \{0.01, 0.05, 0.1\}$ and epochs $\in \{50, 100, 200\}$. Table III summarizes validation F1-Scores for selected configurations.

TABLE III
PERCEPTRON HYPERPARAMETER SEARCH — F1-SCORE ON VALIDATION SET

η	Epochs	F1-Val (Standard)	F1-Val (Pocket)
0.01	50	98.41%	98.41%
0.01	100	98.41%	100.00%
0.01	200	98.41%	98.41%
0.05	50	96.97%	98.41%
0.05	100	98.41%	100.00%
0.05	200	98.41%	98.41%
0.10	50	92.75%	98.41%
0.10	100	95.38%	98.41%
0.10	200	94.12%	**98.41%**

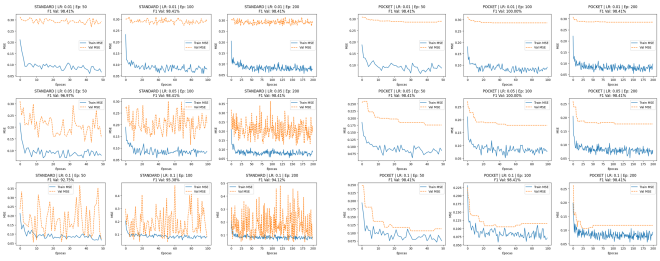

Fig. 4. Validation F1-Score comparison between Standard and Pocket Perceptron across hyperparameter configurations.

TABLE IV
PERCEPTRON FINAL TEST RESULTS

Metric	Standard	Pocket
Accuracy	89.11%	**95.05%**
Precision	76.92%	**93.55%**
Recall	93.75%	90.62%
F1-Score	84.51%	**92.06%**

C. Results and Analysis

The confusion matrices on the test set were: Standard (TN=60, FP=9, FN=2, TP=30) and Pocket (TN=67, FP=2, FN=3, TP=29). The Pocket algorithm's strategy of preserving the historically best weight vector prevents oscillations caused by non-linearly separable data near the decision boundary. This reduced false positives from 9 to 2, a critical improvement in a clinical context where unnecessary interventions carry significant patient impact. The standard Perceptron showed high sensitivity to epochs (F1-Val ranging from 92.75% to 98.41%), while the Pocket version consistently achieved F1-Val $\geq$ 98.41% across all configurations.

III. MLP: MULTILAYER PERCEPTRON WITH BACKPROPAGATION

A. Methodology

A Multilayer Perceptron was implemented from scratch using the Backpropagation algorithm [4], with Sigmoid activation functions and online weight updates. Three key design decisions were made: (1) class imbalance was addressed with a weighted loss function (pos_weight=2.15, computed as benign/malignant ratio in the training set); (2) an adaptive learning rate using the Bold Driver heuristic [6] was implemented, which increases μ by a factor of 1.05 when validation loss decreases and reduces it by 0.7 with weight reversion when loss increases; and (3) Early Stopping with patience=20 epochs was applied on validation loss.

Architecture selection was performed in two sequential steps: first optimizing depth (1–4 hidden layers with 4 neurons each), then optimizing width (4, 8, 16, 32, 64 neurons) at the optimal depth.

B. Architecture Search

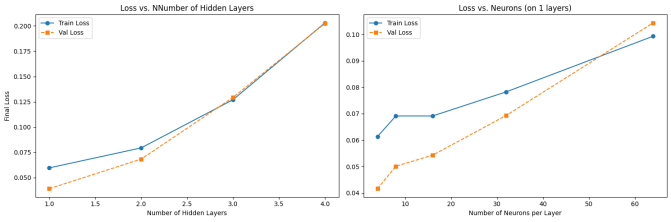

Fig. 5. MLP architecture search: validation loss as a function of number of hidden layers (left) and neurons per layer (right).

The search confirmed that a single hidden layer with 4 neurons ([Input–4–1]) minimized validation loss. Deeper architectures showed no improvement and introduced instability. This result is consistent with the universal approximation theorem [7], which guarantees that a single hidden layer is sufficient for smooth function approximation given adequate neuron count.

C. Results and Analysis

The best model was retrained for up to 3000 epochs, triggering Early Stopping at epoch 1768. The Bold Driver increased the learning rate from an initial value of 0.01 to approximately 4.8 before validation loss degraded, at which point weights were reverted.

TABLE V
MLP FINAL TEST RESULTS

Metric	Value
Accuracy	97.03%
Precision	100.00%
Recall	90.62%
F1-Score	95.08%

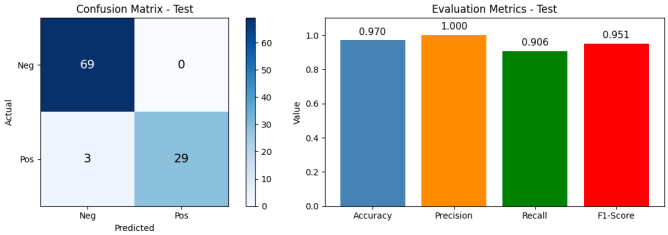

Fig. 6. MLP final evaluation on the test set: confusion matrix (left) and performance metrics (right).

The confusion matrix was: TN=69, FP=0, FN=3, TP=29. The model achieved perfect Precision with only 3 missed malignant cases. The weighted loss function with pos_weight=2.15 successfully mitigated the class imbalance by penalizing false negatives more heavily. Correcting the δ computation at the output layer to include the Sigmoid derivative $z(1-z)$ was essential for proper gradient flow through the network.

IV. CNN ON BINARY MASKS: MORPHOLOGICAL CLASSIFICATION

A. Methodology

A custom CNN was designed to classify tumors based on the binary mask images. A crop-to-mask preprocessing pipeline was implemented: the bounding box of non-zero pixels was extracted, 5px padding was added, and images were resized to 128×128 pixels. This Region of Interest (ROI) extraction eliminates irrelevant background and focuses the convolutional filters on lesion morphology.

Five architectural configurations were evaluated:

Each block included BatchNormalization and MaxPooling. Weighted Cross Entropy loss was used to address class imbalance.

TABLE VI
CNN MASK — ARCHITECTURAL CONFIGURATIONS EVALUATED

Filters	Kernels	LR	Dropout
[16, 32]	[3, 3]	0.001	0.3
[16, 32]	[3, 3]	0.001	0.5
[16, 32, 64]	[3, 3, 3]	0.001	0.3
[16, 32, 64]	[3, 3, 3]	0.001	0.5
[32, 64]	[5, 5]	0.0005	0.3

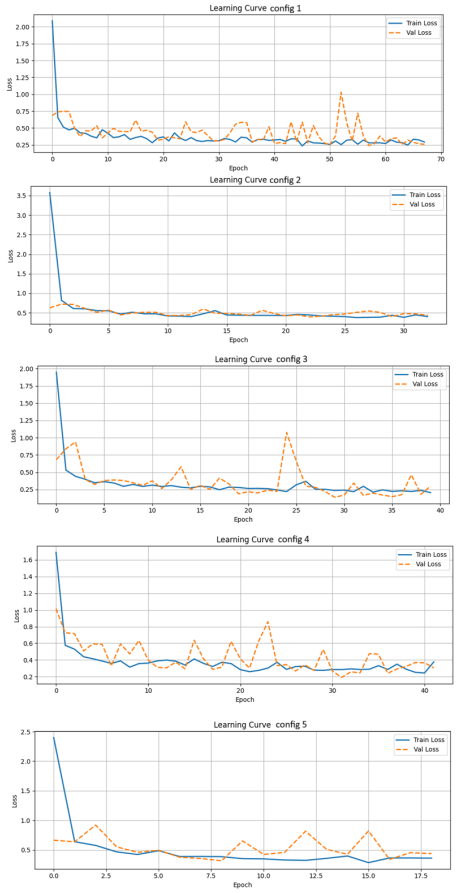

Fig. 7. Training and validation loss for each CNN configuration.

B. Results and Analysis

The best configuration was filters=[16, 32, 64], LR=0.001, Dropout=0.3, achieving F1=0.9677 and Accuracy=0.98 on the test set. The three-block hierarchy successfully captured features at increasing abstraction levels: edges and boundaries in early layers, and complex spiculation patterns characteristic of malignant lesions in deeper layers. Combining LR=0.001 with Dropout=0.3 and BatchNorm provided sufficient regularization to prevent overfitting without restricting model capacity. Dropout=0.5 degraded performance by over-regularizing the relatively small architecture.

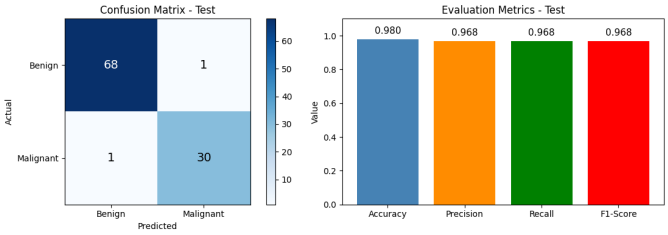

Fig. 8. Final evaluation metrics for the best model.

V. CNN ON RAW ULTRASOUND: TRANSFER LEARNING

A. Methodology

ResNet18 [13] pre-trained on ImageNet was adapted for binary classification of raw ultrasound images. Since ultrasound images are grayscale (1 channel) while ResNet expects RGB (3 channels), the single channel was replicated three times (`repeat(3,1,1)`) to preserve pre-trained weights. Layers `layer1–layer3` were frozen and only `layer4` and the fully connected head were fine-tuned.

Six configurations were evaluated combining two backbones (ResNet18, ResNet34), two learning rates (0.001, 0.0005), and two dropout values (0.3, 0.5).

TABLE VII
RESNET CONFIGURATIONS EVALUATED

Backbone	LR	Dropout	F1-Val
ResNet18	0.001	0.3	73.8%
ResNet18	0.001	0.5	71.2%
ResNet18	0.0005	0.3	69.2%
ResNet18	0.0005	0.5	**78.1%**
ResNet34	0.001	0.3	75.4%
ResNet34	0.0005	0.5	74.2%

B. Results and Analysis

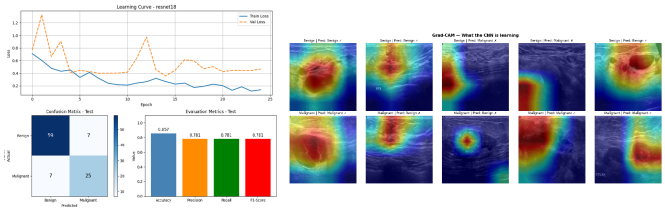

Fig. 9. Training and validation loss, and metrics for the best ResNet18 configuration (left) and Grad-CAM visualization showing network attention regions (right).

The best configuration (ResNet18, LR=0.0005, Dropout=0.5) achieved F1=0.7812, with Early Stopping triggered at epoch 26. Grad-CAM visualizations confirmed that the network frequently focused on background speckle noise and peripheral tissue textures rather than the lesion itself. This misalignment arises from the absence of explicit ROI annotations: when multiple tumors appear in a single image, automated bounding box extraction is unreliable,

forcing the model to learn global image statistics instead of lesion-specific features. The replicated grayscale channel preserved ImageNet weights but introduced a distribution mismatch between natural images and medical ultrasound, limiting the effectiveness of the learned representations.

VI. Conclusions

This study evaluated five neural network architectures for binary breast tumor classification, revealing a clear performance hierarchy driven by input representation quality rather than model complexity alone.

The LMS model established that the seven morphological features are largely linearly separable, achieving Precision=100% and Accuracy=93.07% at the cost of 7 false negatives. Its convex error surface rendered hyperparameter choice largely irrelevant, though its limited capacity makes it unsuitable as a clinical screening tool.

The Pocket Perceptron demonstrated that preserving historically optimal weights during non-convex search is critical for non-linearly separable data, reducing false positives from 9 to 2 compared to the standard variant and achieving F1=92.06%.

The MLP confirmed that non-linear boundaries are necessary: by correcting gradient computation, addressing class imbalance (pos_weight=2.15), and applying the Bold Driver heuristic, a minimal [Input–4–1] architecture achieved F1=95.08% with Precision=100%. Early Stopping at epoch 1768 prevented overfitting without sacrificing generalization.

The Mask CNN achieved the highest overall performance (F1=96.77%), demonstrating that clean morphological information from binary masks — particularly the contrast between spiculated malignant borders and smooth benign contours — is highly discriminative. ROI cropping was the single most impactful preprocessing step.

The raw-image ResNet18 exposed the limitations of transfer learning without explicit tumor localization: Grad-CAM heatmaps revealed attention to irrelevant background regions, and the F1=78.12% ceiling reflects the high noise-to-signal ratio in unprocessed ultrasound. These results suggest that a two-stage pipeline — detection followed by classification — would substantially improve performance on raw images.

References

[1] B. Widrow and M. E. Hoff, "Adaptive switching circuits," *1960 IRE WESCON Convention Record, Part 4*, New York: IRE, pp. 96–104, 1960.

[2] F. Rosenblatt, "The perceptron: A probabilistic model for information storage and organization in the brain," *Psychological Review*, vol. 65, no. 6, pp. 386–408, 1958.

[3] S. I. Gallant, "Perceptron-based learning algorithms," *IEEE Transactions on Neural Networks*, vol. 1, no. 2, pp. 179–191, 1990.

[4] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, pp. 533–536, 1986.

[5] P. J. Werbos, "Beyond regression: New tools for prediction and analysis in the behavioral sciences," Ph.D. dissertation, Harvard University, 1974.

[6] R. A. Jacobs, "Increased rates of convergence through learning rate adaptation," *Neural Networks*, vol. 1, no. 4, pp. 295–307, 1988.

[7] G. Cybenko, "Approximation by superpositions of a sigmoidal function," *Mathematics of Control, Signals and Systems*, vol. 2, no. 4, pp. 303–314, 1989.

[8] K. Hornik, "Approximation capabilities of multilayer feedforward networks," *Neural Networks*, vol. 4, no. 2, pp. 251–257, 1991.

[9] Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel, "Backpropagation applied to handwritten zip code recognition," *Neural Computation*, vol. 1, no. 4, pp. 541–551, 1989.

[10] C. M. Bishop, *Neural Networks for Pattern Recognition*. New York: Oxford University Press, 1995.

[11] C. Lurette, "Développement d'une technique neuronale auto adaptative pour la classification dynamique de données évolutives: Application a la supervision d'une presse hydraulique," Ph.D. dissertation, Université des Sciences et Technologies de Lille, Lille, France, 2003.

[12] J. M. Anguas Pérez, *Análisis y Diseño de Dispositivos de Radiocomunicación Mediante Redes Neuronales*, Ch. 4. Consultado agosto 2021.

[13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, pp. 770–778, 2016.

[14] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, "Dataset of breast ultrasound images," *Data in Brief*, vol. 28, p. 104863, 2020.

[15] C. Isaza, "Fundamentos de Redes Neuronales," *Course Notes, Departamento de Ingeniería Electrónica, Universidad de Antioquia*, Medellín, Colombia, 2026.

[16] J.P. González Blandón, "Breast Cancer Classification with Neural Networks," GitHub, 2026. [Online]. Available: https://github.com/JuanGonzalez47/Breast-Cancer-Classification-with-Neural-Networks